

MODULE 4: ASYMMETRIC KEY ALGORITHM & RSA

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Asymmetric Key Cryptography

Limitations of Symmetric Key Cryptography

1. **Key Distribution Problem:** The secret key must be shared securely between sender and receiver before communication.
2. **Scalability Issue:** The number of required keys increases rapidly as the number of users grows.
3. **Difficult Key Management:** Generating, distributing, storing, and updating keys becomes complex.
4. **Lack of Non-Repudiation:** Since both parties use the same key, the sender cannot be uniquely identified.
5. **Limited Authentication:** Symmetric encryption does not effectively verify user identity.
6. **Single Key Dependency:** The entire communication relies on one shared secret key.
7. **Key Compromise Risk:** If the secret key is exposed, all encrypted data can be accessed.
8. **Secure Storage Requirement:** Keys must be protected from unauthorized access and theft.
9. **High Administrative Overhead:** Managing numerous keys requires significant effort and resources.
10. **Not Suitable for Large Networks:** Key management becomes difficult in organizations with many users.
11. **Risk to Previous Communications:** A compromised key can expose previously encrypted messages.
12. **Dependence on Secure Channels:** A trusted method is required for exchanging secret keys securely.

Asymmetric Key Cryptography

Asymmetric Key Cryptography, also known as **Public Key Cryptography**, is a cryptographic technique that uses **two different but mathematically related keys** for encryption and decryption.

- One key is called the **Public Key** and can be shared openly.
- The other key is called the **Private Key** and is kept secret by the owner.

This method eliminates the need to share a secret key before communication. **Popular Public Key Algorithms:** RSA, Diffie-Hellman

Working Principle

1. The receiver generates a public key and a private key.
2. The public key is shared with everyone.
3. The sender encrypts the message using the receiver's public key.
4. The encrypted message is transmitted over the network.
5. The receiver decrypts the message using their private key.
6. Only the corresponding private key can decrypt data encrypted by the public key.

Public Key and Private Key

1. **Public Key:** A public key is the key that can be freely distributed to anyone.
 - Shared openly.
 - Used for encryption.
 - Used for signature verification.
 - Does not need to be kept secret.
2. **Private Key:** A private key is the key that must remain confidential.
 - Known only to the owner.
 - Used for decryption.

- Used for digital signature creation.
- Must be protected from unauthorized access.

Relationship Between Keys

- Public and private keys are mathematically related.
- Knowing the public key does not reveal the private key.
- Security depends on keeping the private key secret.

Need for Public Key Cryptography

1. Secure key distribution was difficult.
2. Managing a large number of secret keys was complex.
3. Authentication mechanisms were limited.

4. Non-repudiation could not be achieved.
5. Communication over open networks required stronger security mechanisms.

Advantages

1. **Secure Key Sharing:** Public keys can be shared openly.
2. **Authentication:** Verifies user identity.
3. **Digital Signatures:** Supports electronic signatures.
4. **Non-Repudiation:** Sender cannot deny a message.
5. **Better Scalability:** Fewer keys are required.
6. **No Secret Exchange:** No shared key is needed beforehand.
7. **Internet Friendly:** Suitable for public networks.
8. **High Security:** Private key remains secret.
9. **Secure Key Exchange:** Safely exchanges session keys.
10. **Wide Usage:** Used in modern security systems.

6. **Private Key Risk:** Security depends on key secrecy.
7. **Certificate Dependency:** Often requires certificates.
8. **More Storage:** Needs additional memory.
9. **Key Management Issues:** Poor management reduces security.
10. **Costly Operations:** Cryptographic calculations are expensive.

Applications

1. **Secure Email:** Protects email communication.
2. **Digital Signatures:** Verifies documents and messages.
3. **HTTPS Security:** Secures web browsing.
4. **Online Banking:** Protects banking transactions.
5. **E-Commerce:** Secures online payments.
6. **VPN Security:** Enables secure remote access.
7. **Software Verification:** Authenticates software updates.
8. **PKI Systems:** Supports digital certificates.
9. **Cryptocurrency:** Used in blockchain systems.
10. **Key Exchange:** Distributes secret keys securely.

Limitations

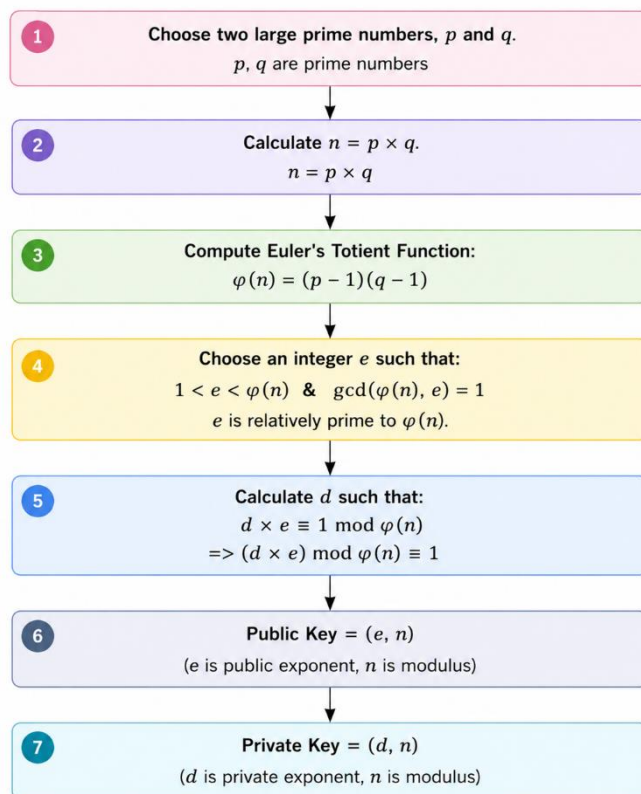
1. **Slower Speed:** Processing is relatively slow.
2. **High Computation:** Requires more resources.
3. **Large Keys:** Uses longer key lengths.
4. **Complex Design:** More difficult to implement.
5. **Lower Efficiency:** Not ideal for large data encryption.

RSA

RSA (Rivest-Shamir-Adleman) is one of the most widely used asymmetric key cryptographic algorithms. RSA uses a pair of keys—a public key for encryption and a private key for decryption. It provides confidentiality, authentication, digital signatures, and secure key exchange.

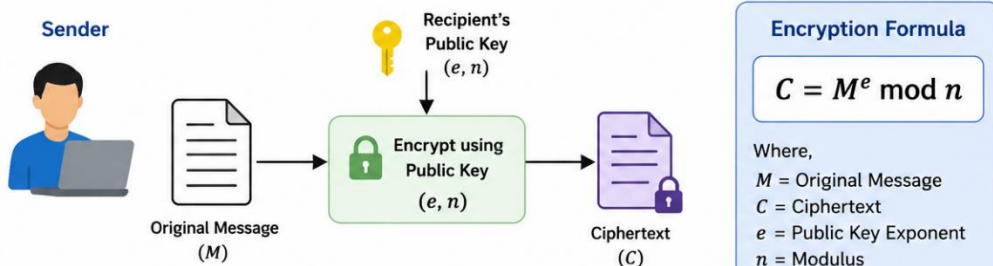
1. Uses public key and private key pair.
2. Based on asymmetric cryptography.
3. Provides confidentiality and authentication.
4. Supports digital signatures.
5. Ensures non-repudiation.
6. Public key can be shared openly.
7. Security depends on factorization of large prime numbers.
8. Widely used in Internet security protocols.
9. Supports secure key exchange.
10. Provides strong security with large key sizes.

RSA Key Generation Process



RSA ENCRYPTION PROCESS

RSA encryption is performed using the recipient's public key.



Where:

C = Ciphertext

M = Original Message

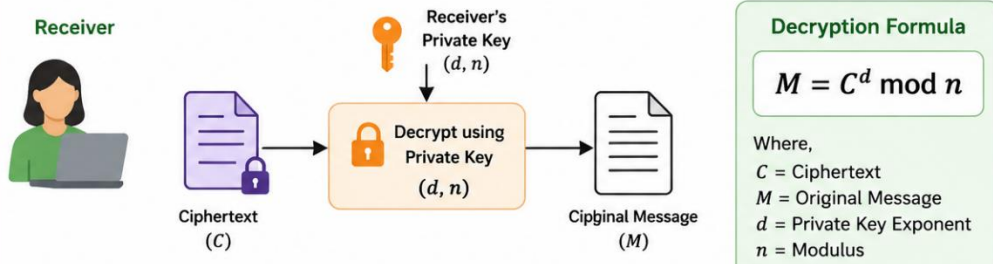
d = Private Key Exponent

n = Modulus

e = Public Key Exponent

RSA DECRYPTION PROCESS

RSA decryption is performed using the receiver's private key.



Example

- ① Generate a private & public key using the prime nos $p=3, q=11$.
Choose $e=7$.
- ② Encrypt the plaintext $M=5$
- ③ Decrypt the ciphertext C (from ②)

Answer

① $n = pq = 33$

$$\phi(n) = (3-1)(11-1) = 20$$

Given, $e = 7$ we check $1 < e < \phi(n)$
 $\Rightarrow 1 < 7 < 20$ (valid)

$$\gcd(7, 20) = 1 \text{ (valid)}$$

$$(de) \bmod \phi(n) \equiv 1 \Rightarrow (d \cdot 7) \bmod 20 \equiv 1$$

$$\boxed{d = 3}$$

$\therefore$ Public key: $(e, n) = (7, 33)$
Private key: $(d, n) = (3, 33)$

② Encryption formula: $C = (M^e) \bmod n$
 $(M=5, e=7, n=33)$
 $= 5^7 \bmod 33$
 $\boxed{C = 14}$

} Sender uses the Receiver's public key $(7, 33)$

③ Decryption formula: $M = (C^d) \bmod n$
 $= 14^3 \bmod 33$
 $\boxed{M = 5}$

} Receiver uses their private key $(3, 33)$

Advantages of RSA

1. Secure public key encryption.
2. Supports digital signatures.
3. Enables secure key exchange.
4. Provides authentication.
5. Ensures non-repudiation.
6. Public keys can be shared openly.
7. Widely accepted standard.

8. Suitable for Internet security.

Disadvantages of RSA

1. Slower than symmetric algorithms.
2. Requires large key sizes.
3. Computationally expensive.
4. Not efficient for large data encryption.
5. Complex key management.
6. Higher memory usage.

7. Vulnerable if private key is exposed.
8. Performance decreases with larger keys.

Applications of RSA

1. Secure web communication (HTTPS).
2. SSL/TLS protocols.
3. Digital signatures.
4. Secure email systems.

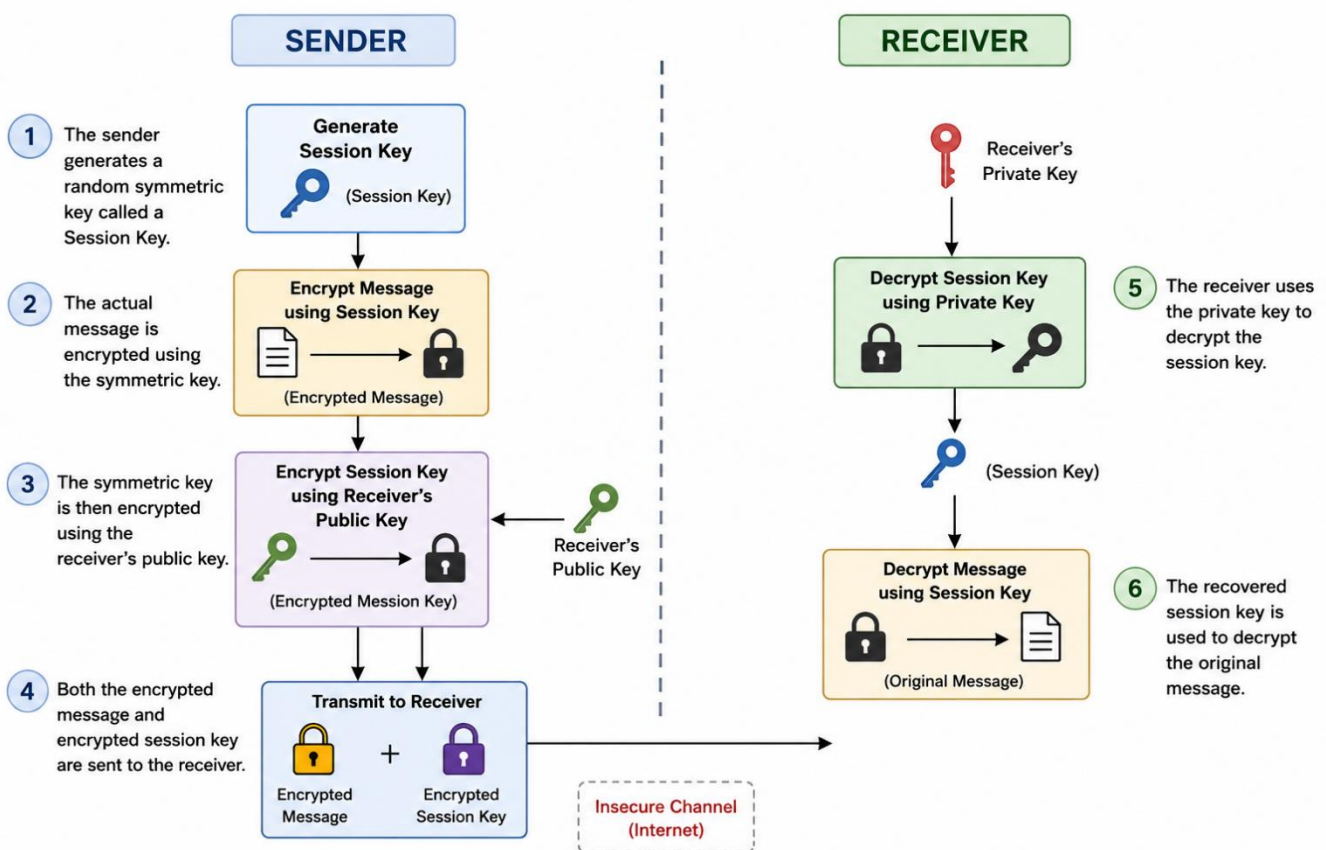
5. Online banking transactions.
6. E-commerce security.
7. Virtual Private Networks (VPNs).
8. Software authentication.
9. Public Key Infrastructure (PKI).
10. Secure exchange of cryptographic keys.

Symmetric and Asymmetric Key Cryptography Together

Hybrid Cryptography is a security approach that combines Symmetric Key Cryptography and Asymmetric Key Cryptography to utilize the advantages of both methods. It uses asymmetric encryption for secure key exchange and symmetric encryption for fast data transmission.

Need for Hybrid Cryptography

1. Symmetric encryption is fast but suffers from key distribution problems.
2. Asymmetric encryption solves key distribution issues but is computationally slower.
3. Hybrid cryptography combines the speed of symmetric encryption with the security of asymmetric encryption.
4. It provides secure communication over insecure networks.
5. It is widely used in modern Internet security systems.



Session Key: is a temporary symmetric key used for encrypting and decrypting data during a single communication session.

1. Generated randomly for each session.
2. Used only for a limited period.
3. Provides fast encryption and decryption.
4. Enhances security by reducing key reuse.
5. Discarded after the communication session ends.

Importance of Session Key

1. Improves encryption speed.
2. Reduces computational overhead.
3. Enhances confidentiality of transmitted data.
4. Minimizes the impact of key compromise.
5. Enables secure communication in hybrid cryptography.

Examples: SSL/TLS, PGP and S/MIME

1) SSL/TLS: SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are protocols used to secure Internet communications. **Working**

1. Asymmetric encryption is used to authenticate the server and exchange keys.
2. A session key is established securely.
3. Symmetric encryption is used for data transfer.
4. Provides confidentiality, integrity, and authentication.

2) PGP (Pretty Good Privacy): PGP is used for securing emails and files. **Working**

1. A session key is generated.
2. The message is encrypted using the session key.
3. The session key is encrypted using the receiver's public key.
4. The receiver decrypts the session key using the private key and then decrypts the message.

3) S/MIME (Secure/Multipurpose Internet Mail Extensions): S/MIME is a standard for secure email communication. **Working**

1. Uses digital certificates for authentication.
2. Generates a session key for message encryption.
3. Encrypts the session key using public key cryptography.
4. Provides email encryption and digital signatures.

Advantages of Hybrid Cryptography

1. Combines speed and security in a single system.
2. Solves the key distribution problem of symmetric cryptography.
3. Faster than using pure asymmetric encryption.
4. Provides secure communication over public networks.
5. Supports digital signatures and authentication.
6. Suitable for large-scale data transmission.
7. Reduces computational overhead.
8. Enhances confidentiality and integrity.
9. Widely used in modern security protocols.
10. Forms the foundation of HTTPS, VPNs, and secure email systems.

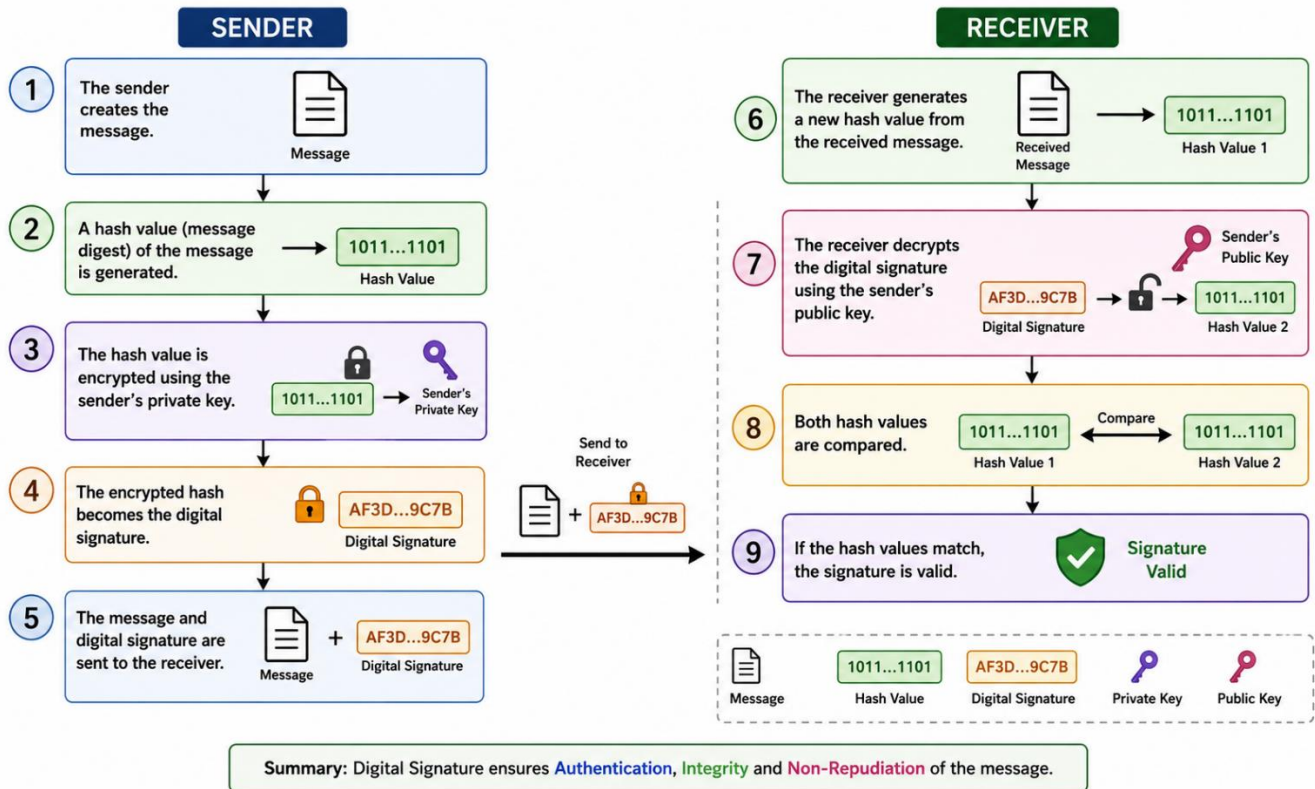
Digital Signature

A Digital Signature is a cryptographic technique used to verify the authenticity, integrity, and origin of a digital message or document. It is the electronic equivalent of a handwritten signature but provides much higher security. Digital signatures use asymmetric key cryptography, where the sender signs the message using a private key and the receiver verifies it using the corresponding public key.

1. Used to authenticate the sender of a message.
2. Ensures data integrity during transmission.
3. Provides non-repudiation.
4. Based on public key cryptography.
5. Widely used in secure electronic transactions.

Objectives of Digital Signature

1. **Authentication:** Verifies the identity of the sender.
2. **Integrity:** Ensures that the message has not been altered.
3. **Non-Repudiation:** Prevents the sender from denying the message.
4. **Security:** Protects digital communications from forgery.
5. **Trust:** Establishes confidence in electronic transactions.



Benefits of Digital Signature

1. Provides strong authentication.
2. Ensures message integrity.
3. Prevents message tampering.
4. Supports non-repudiation.
5. Reduces fraud in electronic transactions.
6. Enables secure online communication.
7. Eliminates the need for physical signatures.
8. Saves time and paperwork.
9. Legally accepted in many countries.
10. Improves trust in digital systems.

Applications of Digital Signature

1. Electronic document signing.
2. E-Governance services.
3. Online banking transactions.
4. E-Commerce applications.
5. Income tax and GST filing.
6. Software distribution and code signing.
7. Secure email communication.
8. Digital certificates and PKI.
9. Electronic contracts and agreements.
10. Online business transactions.

Message Digest

A Message Digest is a fixed-length value generated from a message using a hash function. It acts as a unique digital fingerprint of the original data. Even a small change in the message results in a completely different digest. Message digests are widely used in digital signatures, authentication systems, and data integrity verification. **Key Points**

1. Generated using a hash function.
2. Represents the original message in condensed form.
3. Has a fixed length regardless of message size.
4. Used to verify data integrity.
5. Forms the basis of digital signatures.

Example

Original Message: "Hello"

Message Digest: A1B2C3D4E5F6... (Example Value)

If the message changes to: "hello"

The generated digest will be completely different.

Characteristics of Message Digest

1. **Fixed Length:** Produces an output of fixed size regardless of input length.
2. **Unique Representation:** Represents the contents of the message uniquely.
3. **One-Way Function:** Original message cannot be recovered from the digest.
4. **Fast Computation:** Can be generated quickly even for large files.
5. **Sensitive to Changes:** Small changes produce completely different digests.
6. **Integrity Checking:** Helps detect unauthorized modifications.
7. **Deterministic:** Same input always produces the same digest.
8. **Collision Resistant:** Difficult to find two messages with the same digest.
9. **Compact:** Requires less storage than the original message.

10. **Secure:** Supports authentication and verification mechanisms.

Uses of Message Digest

1. Verifying data integrity during transmission.
2. Creating digital signatures.
3. Password storage and authentication.
4. Detecting file tampering.
5. Secure email communication.
6. Software and document verification.
7. Electronic transactions and e-commerce.
8. Blockchain and cryptocurrency systems.
9. Certificate and security protocols.
10. Authentication mechanisms in computer networks.

Hash Function

A Hash Function is a mathematical algorithm that converts data of any size into a fixed-length value known as a hash value or message digest. It plays a vital role in modern cryptography by ensuring data integrity, authentication, and security. Hash functions are widely used in digital signatures, password protection, blockchain technology, and secure communication systems. **Key Points**

1. Converts input data into a fixed-size output.
2. Produces a unique digital fingerprint of data.
3. Helps detect unauthorized modifications.
4. Used extensively in cryptographic applications.
5. Supports digital signatures and authentication mechanisms.

Formula

$$H(M) = D$$

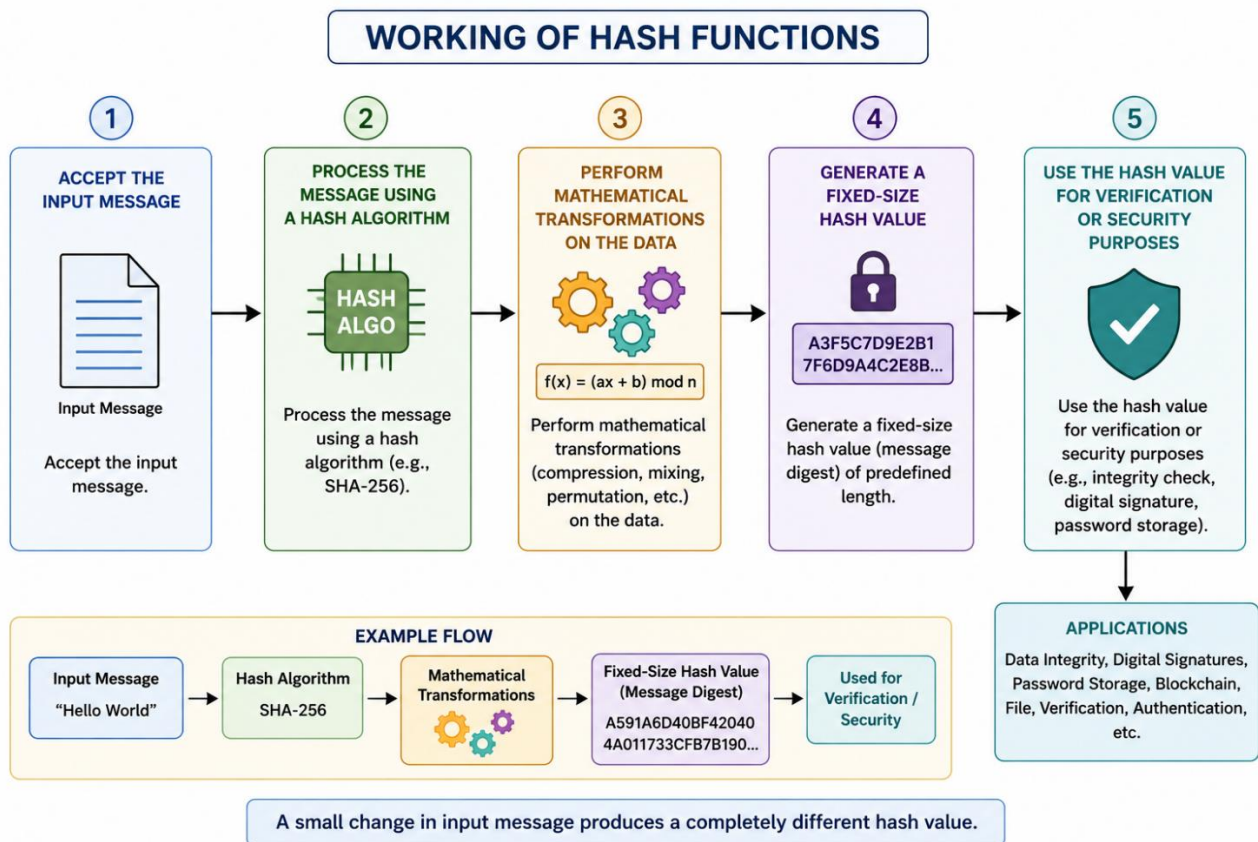
Where:

- H = Hash Function
- M = Input Message
- D = Hash Value (Message Digest)

Properties of a Good Hash Function

1. **Fixed-Length Output:** Produces a hash value of fixed size regardless of input size.
2. **Deterministic:** The same input always generates the same hash value.
3. **Fast Computation:** Hash values can be generated quickly and efficiently.
4. **One-Way Property (Pre-image Resistance):** It should be computationally impossible to obtain the original message from the hash value.
5. **Collision Resistance:** It should be extremely difficult to find two different messages that generate the same hash value.

6. **Avalanche Effect:** A small change in the input should produce a completely different hash value.
7. **Uniform Distribution:** Hash values should be distributed uniformly to minimize collisions.
8. **Second Pre-image Resistance:** Given a message and its hash, it should be difficult to find another message with the same hash.



Applications of Hash Functions

1. **Data Integrity Verification:** Used to verify whether data has been modified during transmission or storage.
2. **Digital Signatures:** Hash values are signed instead of entire messages to improve efficiency.
3. **Password Storage:** Passwords are stored as hashes rather than plain text.
4. **File Tampering Detection:** Used to detect unauthorized changes in files and documents.
5. **Secure Email Communication:** Ensures message integrity and authenticity.
6. **Blockchain Technology:** Each block contains hashes linking it to previous blocks.
7. **Software Verification:** Used to verify downloaded software and updates.
8. **Database Indexing:** Helps in fast data retrieval and searching.
9. **Authentication Systems:** Used in login and identity verification processes.
10. **Cryptographic Protocols:** Forms a fundamental component of SSL/TLS, VPNs, and security protocols.

Aspect	Hash Function	Message Digest
Definition	Algorithm used to generate a hash value	Output produced by a hash function
Nature	Process / Mathematical Function	Result / Output
Input	Accepts data of any length	Not applicable
Output	Produces a message digest	Fixed-length hash value
Purpose	To calculate a hash value	To represent the original data uniquely
Example	SHA-256, SHA-512, MD5, SHA-1	A591A6D40BF42040A011733CFB7B190...
Reversibility	One-way function	Cannot reveal original message

Change Input	in	Produces a different digest	Represents changed data
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Key Difference

- **Hash Function** is the algorithm or method used to process data.
- **Message Digest** is the fixed-length output generated by the hash function.

Example

Input Message



Hash Function (SHA-256)



Message Digest

Thus, the hash function is the process, while the message digest is the result produced by that process.